SG3525A PWM Controller

Mode-1 : Open-Loop, Fixed Duty Cycle (50 %) at 10 kHz

LTspice simulation · circuit design, working and waveform analysis

1. Objective

To operate the SG3525A PWM controller in **Mode-1**: the control loop is open, there is no feedback from the system output, and the IC simply produces a **fixed duty cycle (≈ 50 %) square wave at f_{out} = 10 kHz** on output A (pin 11). The oscillator runs at f_{osc} = 20 kHz and the internal flip-flop divides it by two.

2. Why Mode-1 behaves this way (concept)

The SG3525A compares the output of its error amplifier with the oscillator ramp on pin 5 (C_T). The width of the output pulse is decided by where the error-amplifier voltage cuts the ramp. In Mode-1 we deliberately **force the error amplifier to saturate high** (non-inverting input above inverting input). The amplifier output then sits above the top of the ramp and never cuts it, so the PWM comparator never ends the pulse early. The pulse width is therefore set only by the oscillator itself – it is fixed, independent of any system output.

3. Circuit diagram

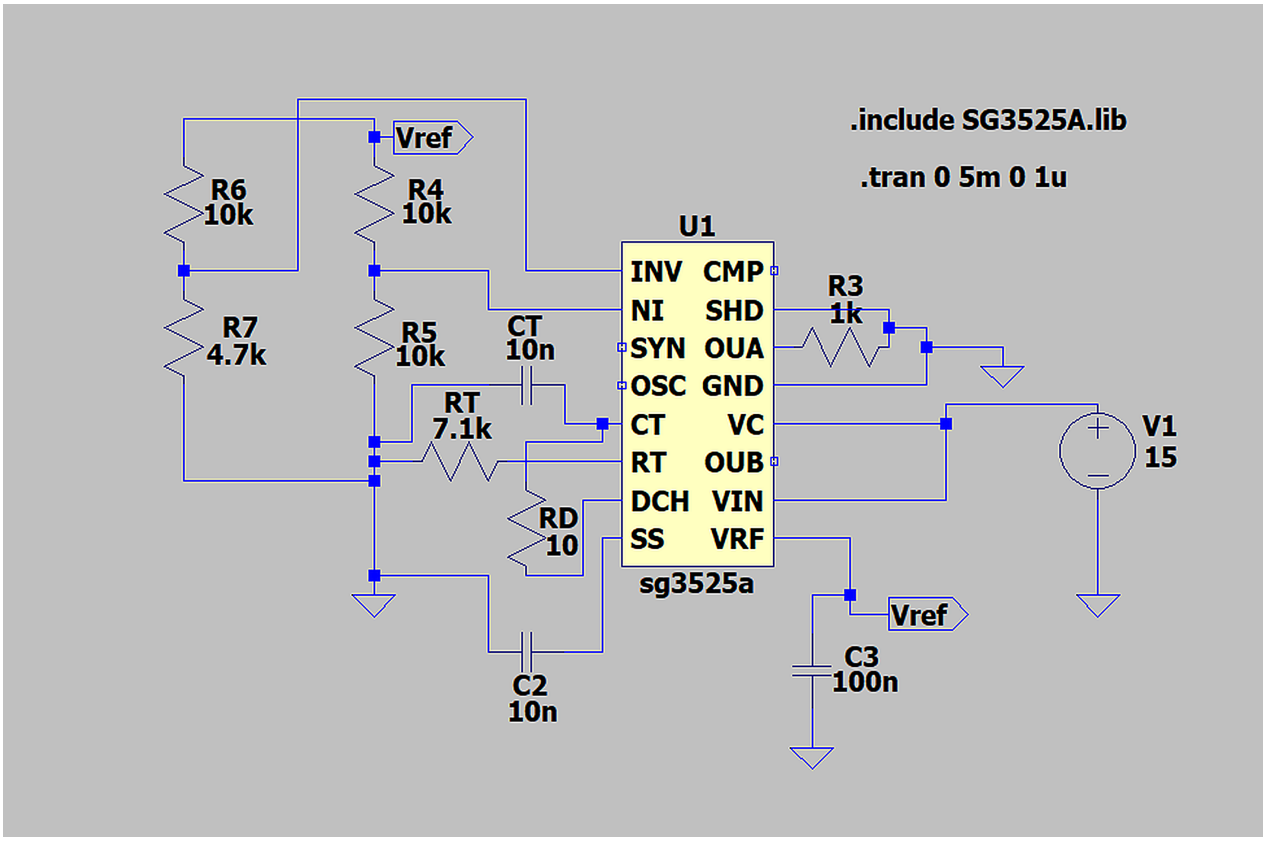


Fig. 1 – LTspice schematic of SG3525A in Mode-1 (V1 = 15 V supply, .tran 0 5m 0 1u).

4. Pin connections and why each is made

The table follows the notebook analysis. Each pin is connected the way it is for a specific reason.

Pin	Name	Connection	Why
15	V_{IN}	+15 V (V1)	Supply of the IC; powers the internal 5.1 V regulator and logic.
13	V_C	+15 V (same as V_{IN})	Supply of the totem-pole output stage; sets the output high level.
12	GND	Ground	Common reference.
16	V_{REF}	5.1 V internal regulator output, 100 nF to GND	Reference for the divider networks; 100 nF decoupling keeps it stable.
1	INV	Divider R6 (10 kΩ) / R7 (4.7 kΩ) from V_{ref}	Gives a fixed ≈ 1.63 V at the inverting input.
2	N-INV	Divider R4 (10 kΩ) / R5 (10 kΩ) from V_{ref}	Gives a fixed ≈ 2.56 V at the non-inverting input. Since 2.56 V > 1.63 V, the error amplifier saturates high → fixed duty.
5	C_T	10 nF to GND	Timing capacitor – produces the sawtooth ramp.
6	R_T	7.1 kΩ to GND	Timing resistor – sets the charging current of C_T.
7	DISCHARGE	10 Ω (R_D) to pin 5	Sets the discharge speed of C_T, i.e. the dead time.
8	SOFT-START	10 nF (C2) to GND	Slows the build-up of duty cycle at start-up (soft start).
9	COMP	Not connected	No external compensation / feedback in open-loop mode.
10	SHUTDOWN	Ground	Keeps the IC enabled (shutdown inactive).
11	OUT A	R3 = 1 kΩ to GND	Load / probe point for the output waveform.
14	OUT B	Not connected	Only one output is used.
3, 4	SYNC, OSC OUT	Not connected	No external synchronisation needed.

5. Component values and design calculations

Oscillator frequency (notebook formula, with C_T = 10 nF, R_T = 7.1 kΩ, R_D = 10 Ω):

> $f_{osc} = 1 / [\, C_T (0.7\, R_T + 3\, R_D) \,] = 1 / [\, 10\text{ nF} \times (0.7 \times 7.1\text{ k}\Omega + 3 \times 10\ \Omega) \,] = 1 / (10\text{ nF} \times 5\text{ k}\Omega) =$ **20 kHz**

Output frequency: the internal flip-flop alternates between output A and B on each oscillator cycle, so $f_{out} = f_{osc} / 2 =$ **10 kHz**.

Divider voltages (V_{ref} ≈ 5.12 V from the simulation):

> Pin 1 (INV) = $V_{ref} \times 4.7\text{ k} / (10\text{ k} + 4.7\text{ k}) \approx 5.12 \times 0.32 \approx$ **1.64 V**
> Pin 2 (N-INV) = $V_{ref} \times 10\text{ k} / (10\text{ k} + 10\text{ k}) \approx$ **2.56 V**

6. Working – signal flow

- **Power-up:** 15 V on pins 15 and 13. The internal regulator produces ≈ 5.1 V on pin 16 (V_{ref}).
- **Oscillator:** R_T charges C_T from ≈ 0.8 V up to ≈ 3.2 V (sawtooth on pin 5). When it reaches the upper threshold, it is discharged quickly through R_D via pin 7. This repeats at 20 kHz.
- **Error amplifier:** pin 2 (2.56 V) > pin 1 (1.63 V), so the amplifier output saturates high, above the ramp peak.
- **PWM comparator:** because the amplifier output is always above the ramp, the pulse is never terminated early – duty cycle is the maximum allowed by the oscillator (≈ 50 % per output minus dead time).
- **Soft start:** C2 on pin 8 charges slowly and finally rises to ≈ 5.07 V, after which it no longer limits the duty cycle.
- **Output stage:** a flip-flop steers alternate pulses to outputs A and B. Output A (pin 11) therefore delivers a 10 kHz, ≈ 50 % square wave switching between 0 V and ≈ 13.8 V (V_C minus the totem-pole drop).

7. Simulation results (waveforms)

Waveforms are in the order in which the signal chain works: reference → amplifier inputs → oscillator → soft start → amplifier output → final output.

7.1 Reference voltage – pin 16 (V_{ref})

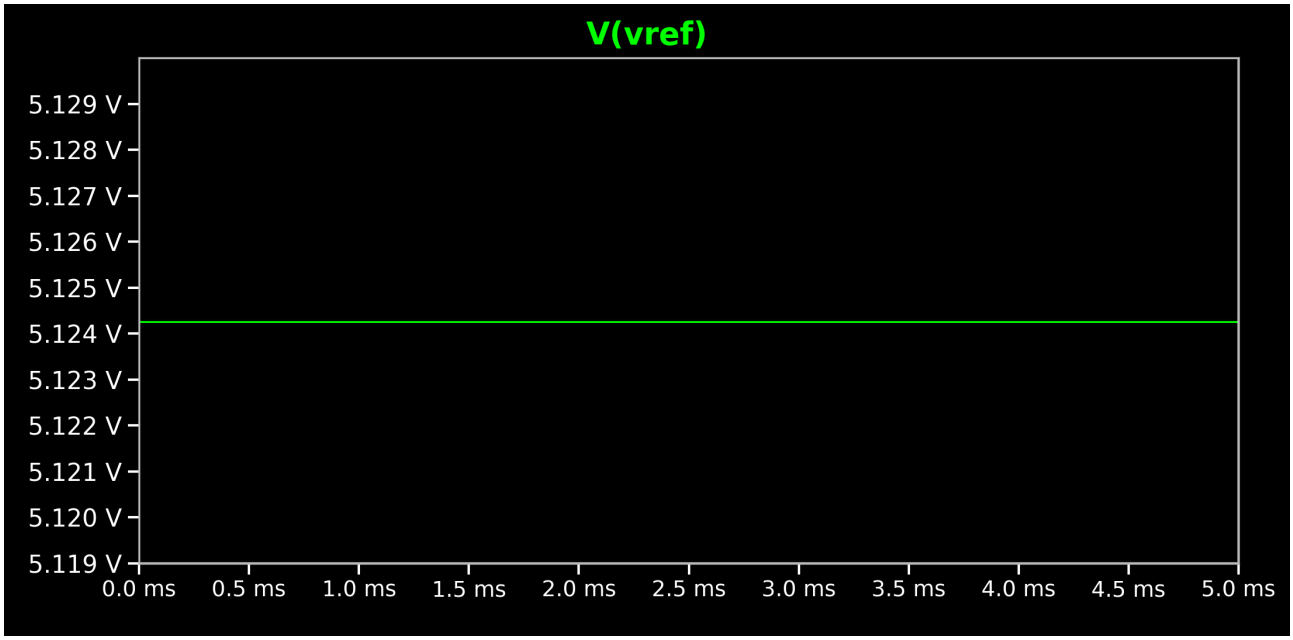


Fig. 2 – V(vref): steady DC at ≈ 5.124 V (datasheet 5.1 V). This is the supply for all the dividers.

7.2 Error-amplifier inputs – pin 1 and pin 2

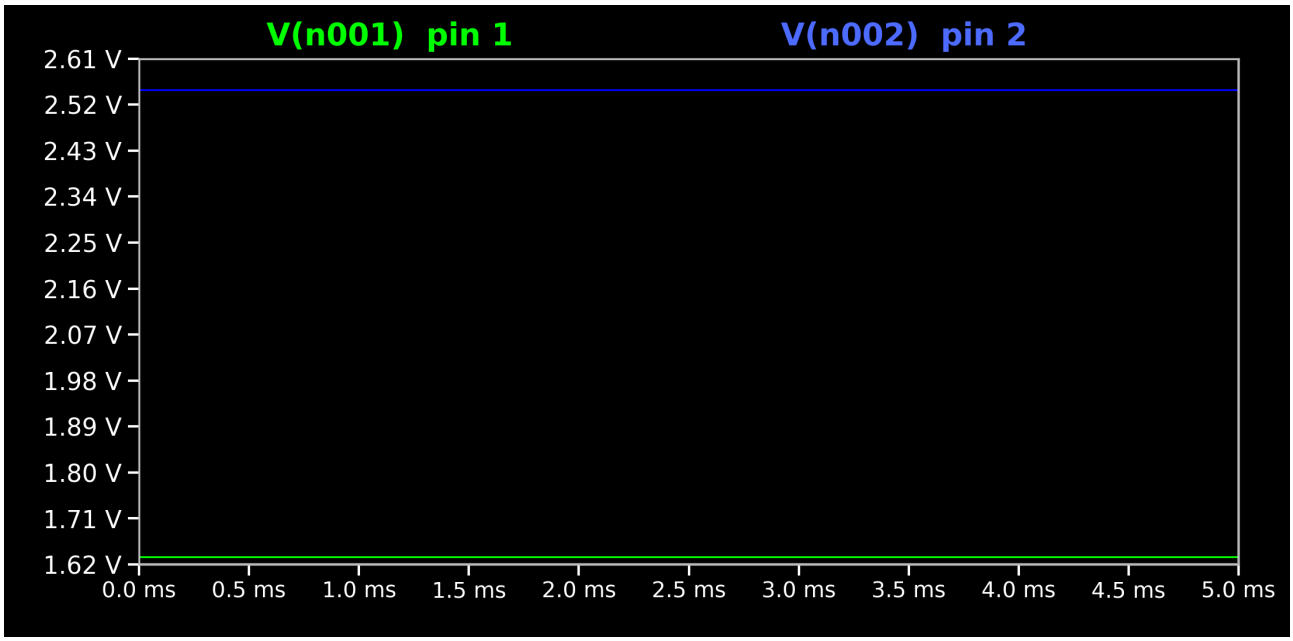


Fig. 3 – V(n001) green = pin 1 (INV) ≈ 1.63 V; V(n002) blue = pin 2 (N-INV) ≈ 2.55 V. Matches the calculated 1.64 V and 2.56 V, and the notebook value of 1.632 V for pin 1.

Pin 2 stays above pin 1 for the whole run, which is the condition that saturates the error amplifier and fixes the duty cycle.

7.3 Oscillator ramp – pin 5 (C_T)

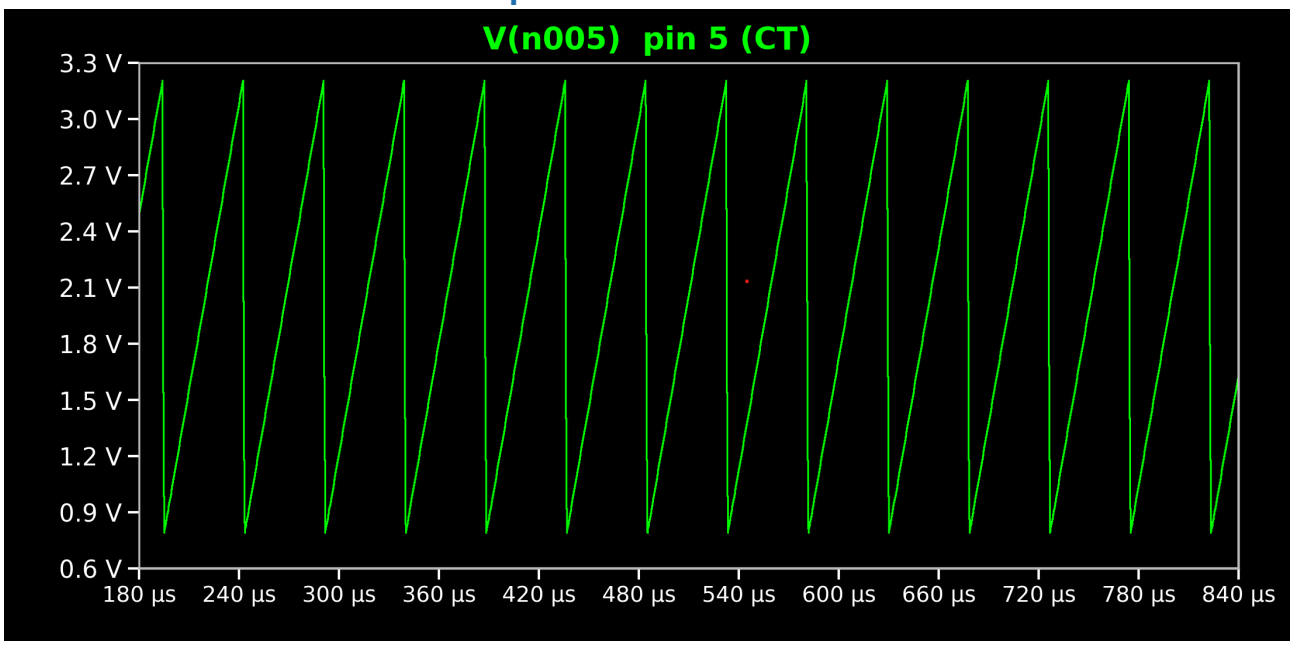


Fig. 4 – V(n005): sawtooth between ≈ 0.8 V and ≈ 3.2 V with a period of ≈ 49 µs, i.e. ≈ 20 kHz, as calculated. Pin 7 follows the same waveform (notebook).

7.4 Soft-start – pin 8

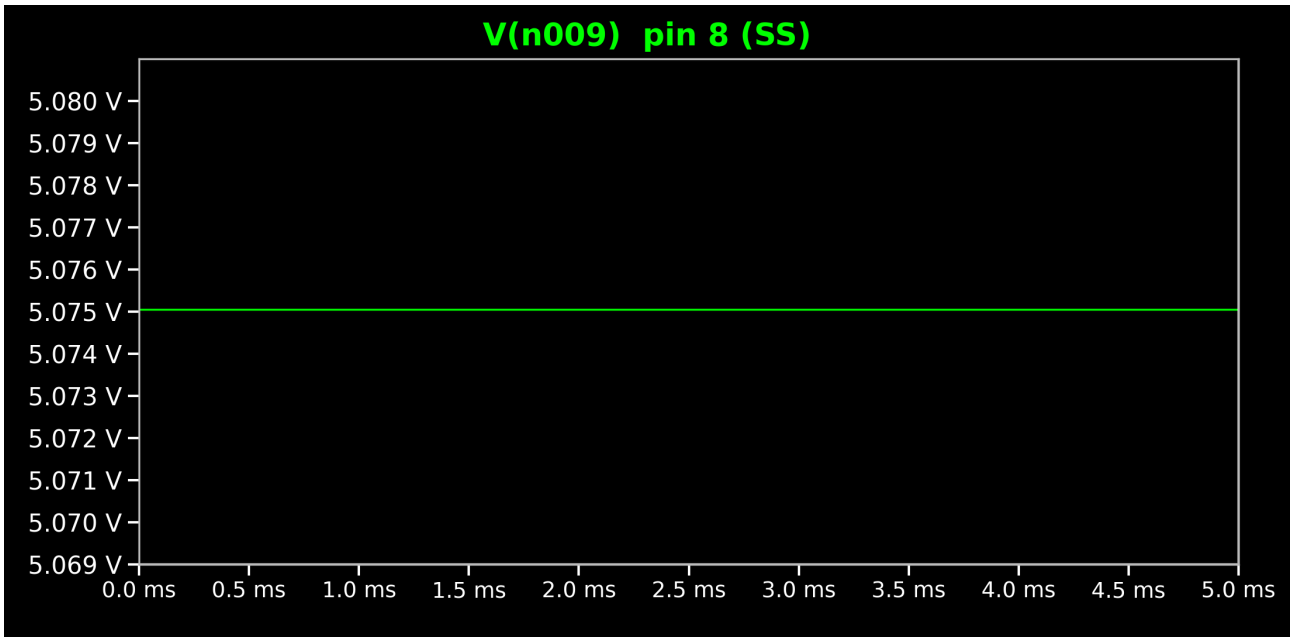


Fig. 5 – V(n009): the soft-start capacitor has settled at ≈ 5.075 V (matches the notebook value for pin 8).

7.5 Error-amplifier output – saturated high

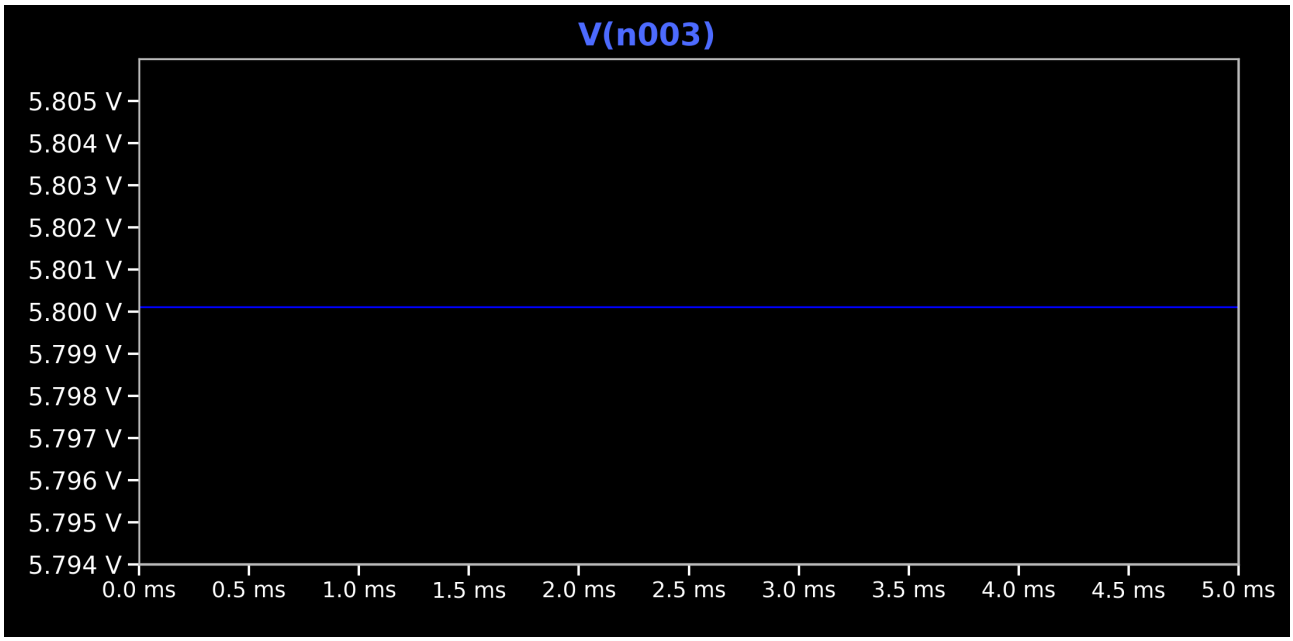


Fig. 6 – V(n003): flat DC at ≈ 5.800 V. This is above the 3.2 V ramp peak, so the PWM comparator never cuts the ramp and the pulse width is fixed.

7.6 Final output – pin 11 (OUT A)

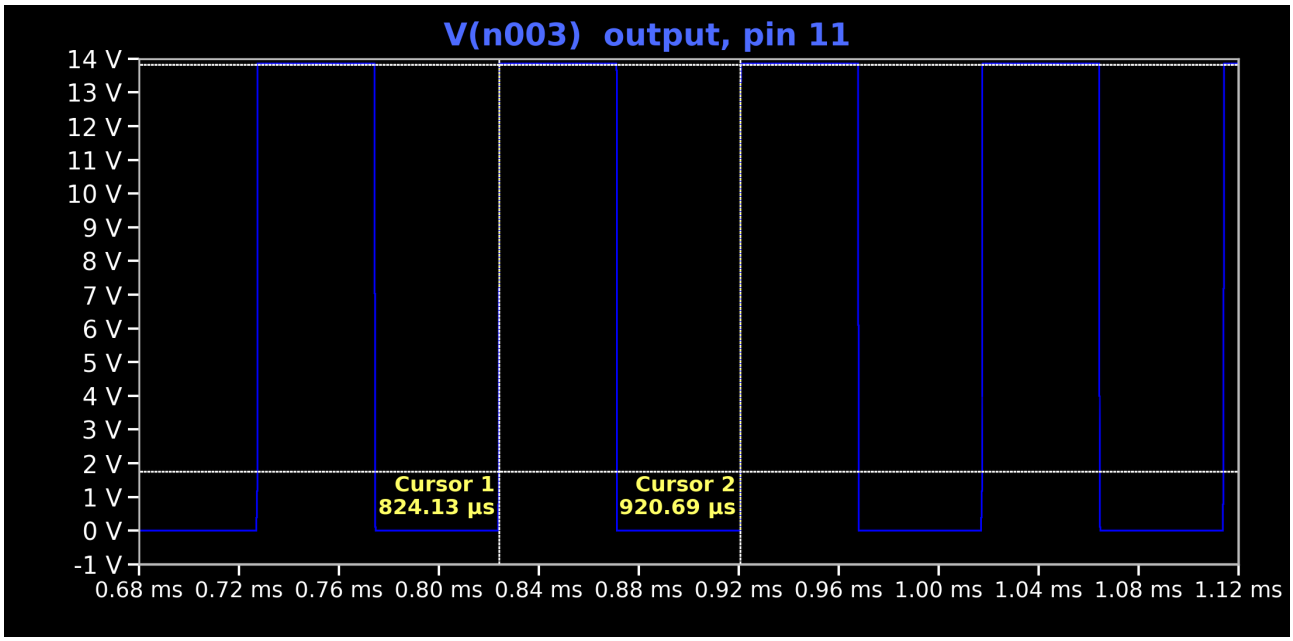
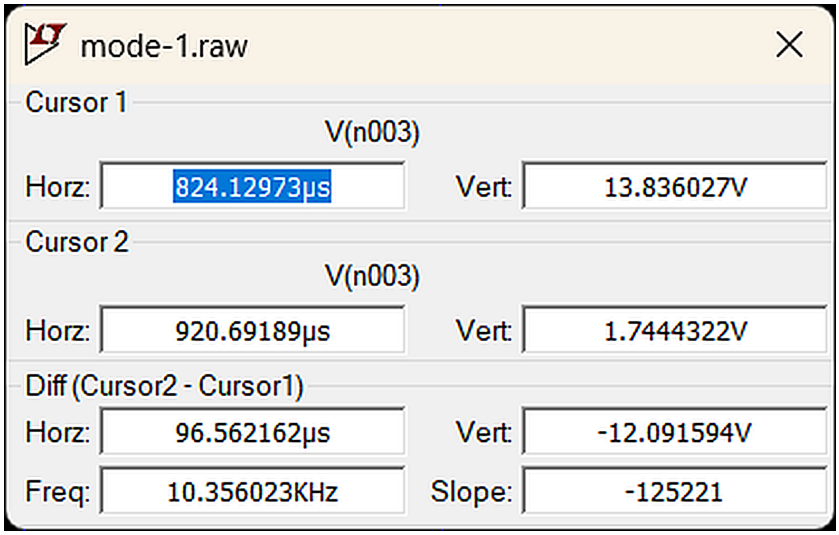


Fig. 7 – Output waveform with cursors: high level 13.84 V, low level 0 V. Cursor difference gives a period of 96.56 µs, i.e. f = 10.356 kHz. High time ≈ 47 µs of 96.6 µs → duty ≈ 48–50 %.



LTspice cursor readout for Fig. 7: Cursor 1 at 824.13 µs (13.836 V), Cursor 2 at 920.69 µs (1.7444 V); difference 96.562 µs, giving Freq = 10.356 kHz.

8. Summary of results

Parameter	Calculated / expected	Simulated
V_{ref} (pin 16)	5.1 V	5.124 V
Pin 1 (INV)	1.64 V	≈ 1.63 V
Pin 2 (N-INV)	2.56 V	≈ 2.55 V
Ramp on C_T (pin 5)	≈ 0.8 V to 3.2 V, 20 kHz	0.8 V to 3.2 V, ≈ 20 kHz
Soft-start (pin 8)	charges to ≈ 5 V	≈ 5.075 V
Error-amp output	saturated high (> 3.2 V)	≈ 5.8 V
Output frequency (pin 11)	10 kHz	10.356 kHz
Output amplitude	≈ V_C (15 V) minus drop	13.84 V high / 0 V low
Duty cycle	≈ 50 %	≈ 48–50 %

9. Conclusion

With pin 2 held above pin 1 by resistor dividers off V_{ref}, the SG3525A's error amplifier saturates and the chip runs open loop. The oscillator set by R_T = 7.1 kΩ and C_T = 10 nF runs at 20 kHz, and the output stage divides this to a fixed ≈ 50 % duty, 10 kHz square wave at pin 11. The simulated voltages and frequencies agree with the hand calculations in the notebook.